1D-SCAPS vs SolarLab 模拟对比报告

Goal: read the SCAPS parameter/result workbook and partner PDF, reproduce feasible tests through SolarLab public APIs, and compare trends one by one.

本报告保持 partner PDF 的顺序：Base Model → band offset → doping → defect density → defect energy → two-dimensional scans。SolarLab 输出均为本地分析文件，不用于 remote push。

Base Model

Layer	Thickness	Eg	chi	eps_r	mu_n	mu_p	N_D	N_A
HTL	20 nm	3.25	2.40	11.75	1e-3	1e-3	0	1e18
PVK	800 nm	1.53	3.94	31	20	20	1e14	0
ETL	25 nm	1.9	4.10	4.2	1e-2	1e-5	1e18	0

Tool	Voc (V)	Jsc (mA/cm2)	FF (%)	PCE (%)	Notes
SCAPS base	1.1676	26.2820	86.99	26.69	from partner PDF
SolarLab SCAPS-like base	1.0020	26.2904	74.68	19.67	Jsc calibrated by photon flux; defect/contact model not yet SCAPS-identical

Capability Summary

Test family	Can SolarLab run now?	Already ran in this report?	Fidelity vs SCAPS
ETL/PVK DeltaEc / chi_ETL	Yes	Yes	Direct chi/Eg mapping, but current response is much flatter than SCAPS
ETL donor doping	Yes	Yes	Direct N_D mapping; FF/PCE trend agrees qualitatively
PVK donor doping	Yes	Yes	Direct N_D mapping; high-doping degradation appears in both
Bulk defect density	Yes, proxy	Yes for PVK-CB proxy	Mapped through SRH lifetime / trap_N_t_bulk
Interface defect density	Yes, proxy	Yes for PVK/ETL edge-trap proxy	Mapped through absorber edge trap profile or SRV
Defect energy Et	Yes, proxy	Yes for CB/VB/PVK-ETL proxy	Mapped through SRH n1/p1; not exact SCAPS interface defect energy
Nt x Et and Nt x DeltaEc 2D	Partly	SCAPS heatmaps included; full SolarLab matrix not yet run	SolarLab runner can do matrices; exact parity requires SCAPS-style interface-defect mapping

ETL/PVK conduction band offset scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
ETL/PVK conduction band offset scan	DeltaEc = -0.16 eV	-1.0 to 0.70 eV	fix chi_PVK=3.94, set chi_ETL=3.94-DeltaEc	direct ETL chi update; V_bi recomputed by SolarLab	Already ran

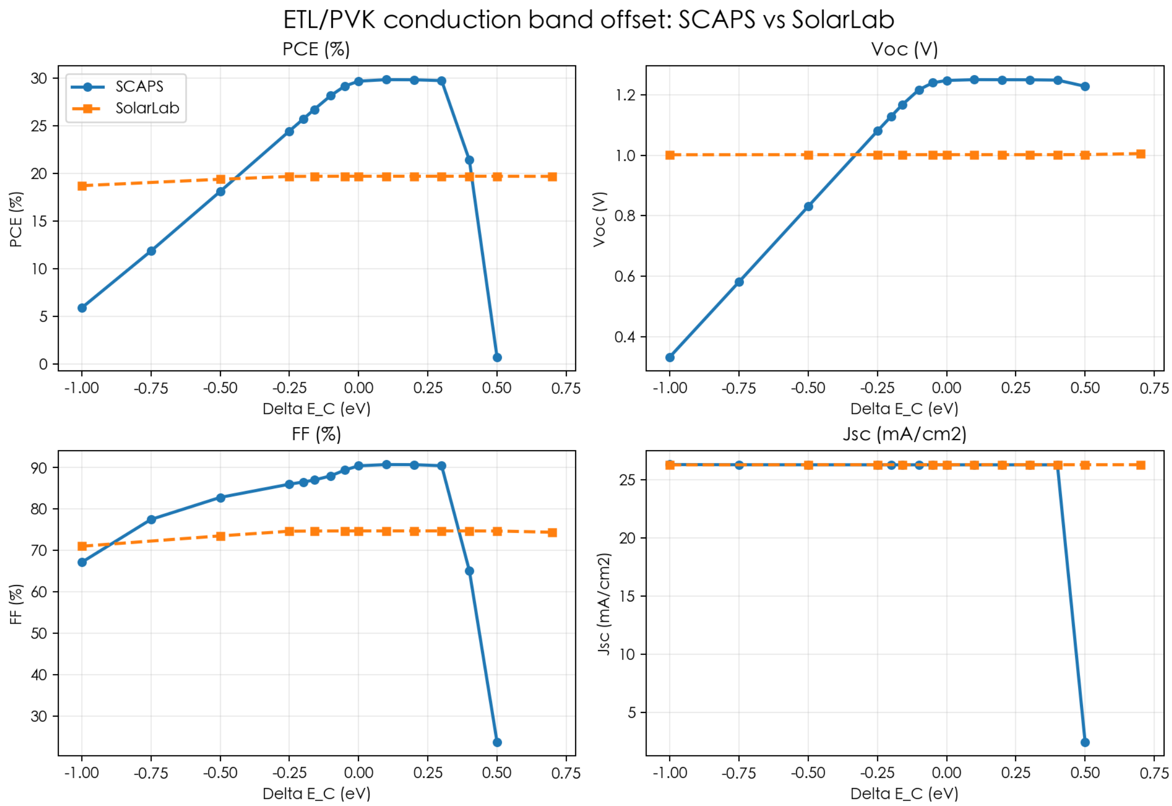
Result record / trend check

Best / base	DeltaEc	PCE	Voc	FF	Jsc
SCAPS base	-0.16	26.69	1.1676	86.99	26.282
SCAPS best parsed	0.10	29.82	1.2504	90.73	26.281
SolarLab base	-0.16	19.67	1.0020	74.68	26.290

Quantitative trend summary

Metric	n	corr	SCAPS range	SolarLab range
PCE (%)	11	0.565	0.7158-29.82	18.69-19.68
Voc (V)	11	0.732	0.3324-1.25	1.002-1.006
FF (%)	11	0.161	23.75-90.73	71-74.7
Jsc (mA/cm2)	11	-0.106	2.453-26.3	26.28-26.29

Visualization



Conclusion

• SCAPS shows a clear optimum window around 0 to 0.3 eV and severe spike loss near 0.5 eV.

• SolarLab successfully runs all sampled points, but the response is too flat in this mapping.

• This is the largest current gap: strict SCAPS usability validation needs calibrated heterointerface barrier / thermionic-interface treatment.

ETL donor doping scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
ETL donor doping scan	1e18 cm^-3	1e10 to 1e20 cm^-3	ETL shallow uniform donor density	direct ETL N_D update	Already ran

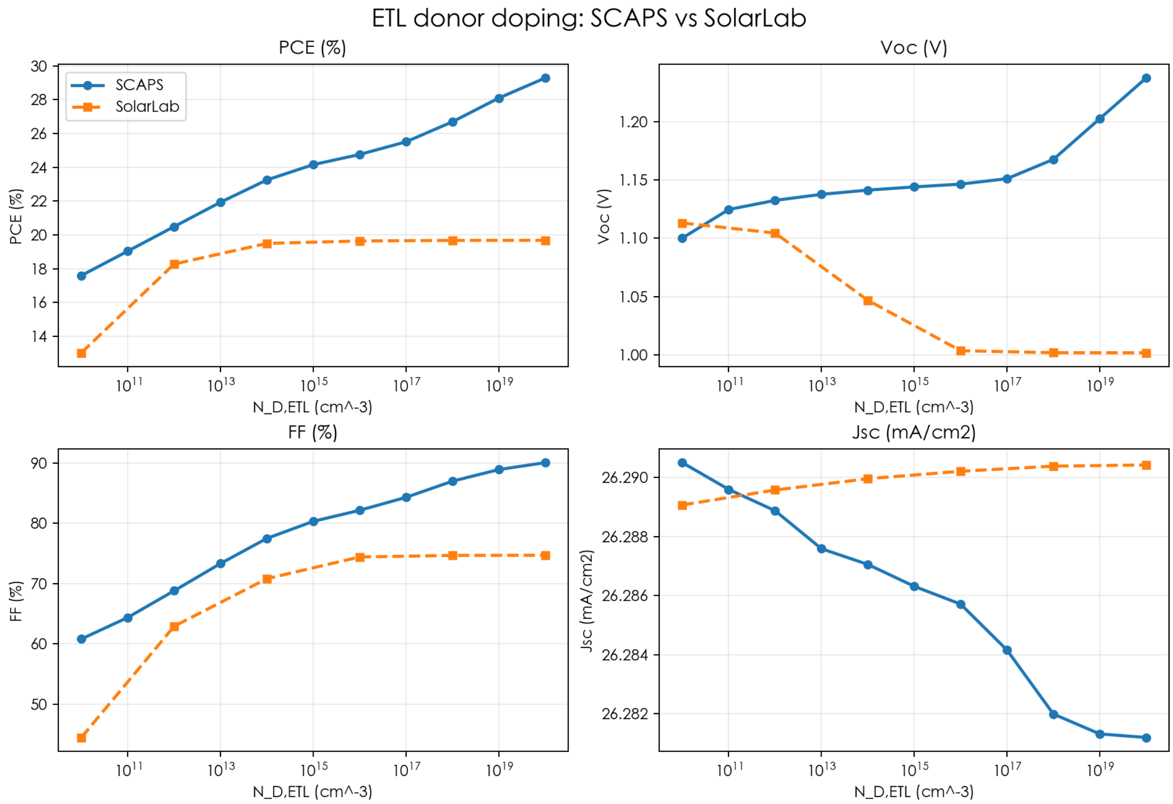
Result record / trend check

Tool	low doping PCE	base PCE	high doping PCE	Main driver
SCAPS	17.59	26.69	29.29	FF and Voc
SolarLab	13.02	19.67	19.68	mostly FF

Quantitative trend summary

Metric	n	corr	SCAPS range	SolarLab range
PCE (%)	6	0.808	17.59-29.29	13.02-19.68
Voc (V)	6	-0.732	1.1-1.237	1.002-1.113
FF (%)	6	0.924	60.83-90.06	44.5-74.7
Jsc (mA/cm2)	6	-0.932	26.28-26.29	26.29-26.29

Visualization



Conclusion

- SCAPS: higher ETL doping improves FF/PCE, Jsc nearly constant.
- SolarLab reproduces the main FF/PCE improvement at low-to-mid doping, then saturates.
- Voc trend differs because SolarLab recomputes built-in potential and does not share SCAPS contact calibration yet.

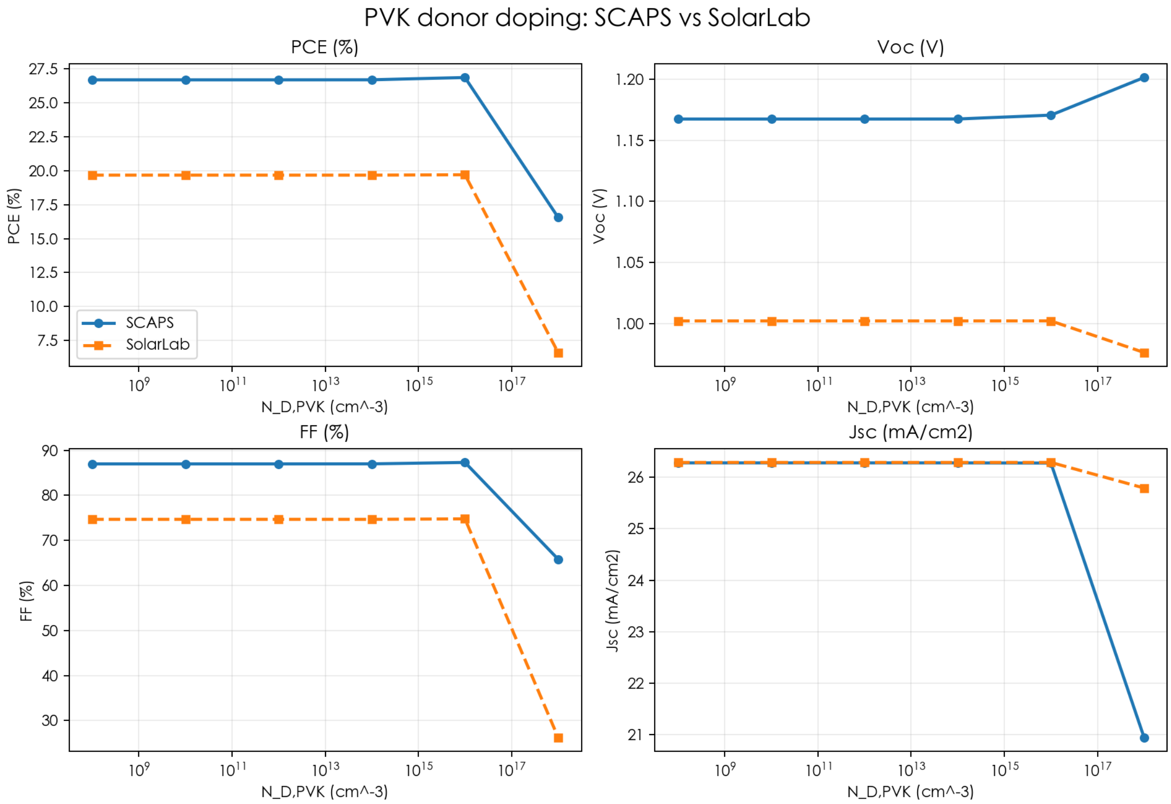
PVK donor doping scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK donor doping scan	1e14 cm^-3	1e8 to 1e18 cm^-3	PVK shallow uniform donor density	direct absorber N_D update	Already ran

Result record / trend check

Tool	stable region	high doping outcome
SCAPS	PCE about 26.7-26.9% through 1e16	drops to 16.57% at 1e18
SolarLab	PCE about 19.67-19.70% through 1e16	drops to 6.61% at 1e18

Visualization



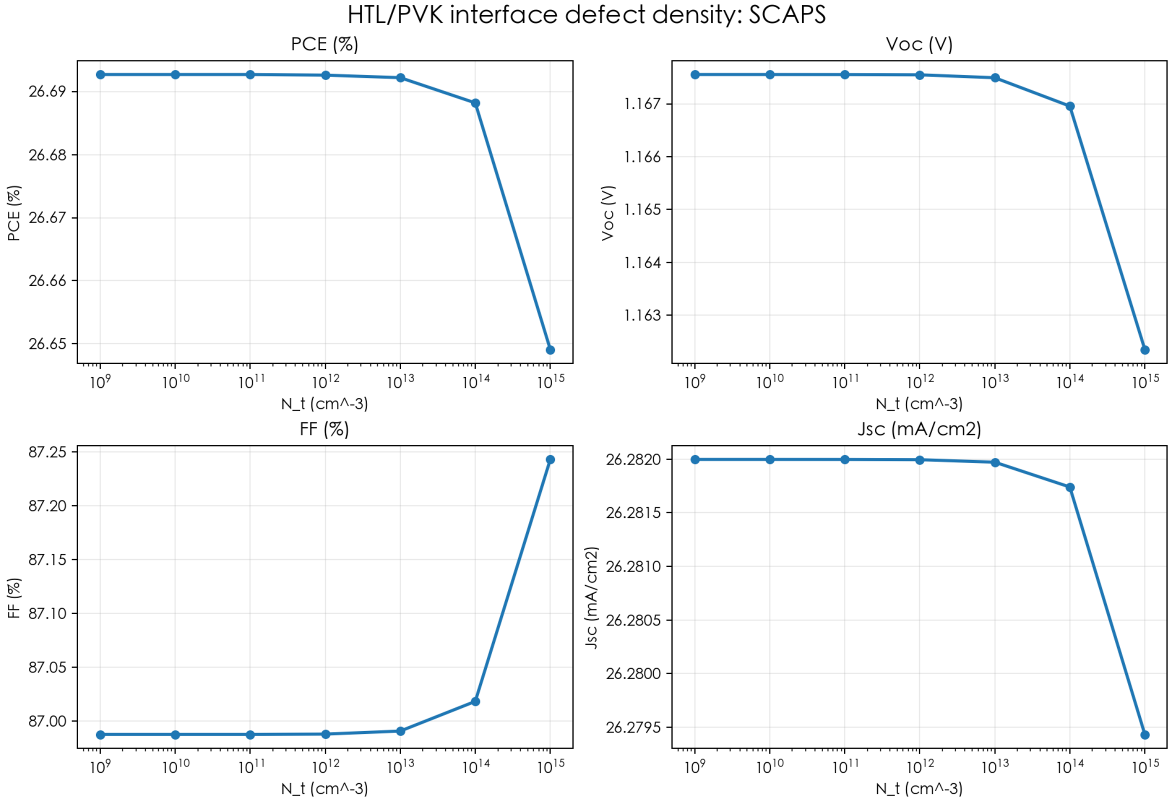
Conclusion

- Both tools are almost flat from low doping through about 1e16 cm^-3.
- Both show strong degradation at 1e18 cm^-3.
- This is a good qualitative usability check for high absorber doping effects.

HTL/PVK interface defect density scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
HTL/PVK interface defect density scan	1e12 cm^-3	1e9 to 1e15 cm^-3	HTL/PVK defect density only	can map through interface SRV or local edge trap; not run in strict parity	Can run; not yet strict

Visualization



Conclusion

• SCAPS indicates HTL/PVK defect density is weakly sensitive in this base model.

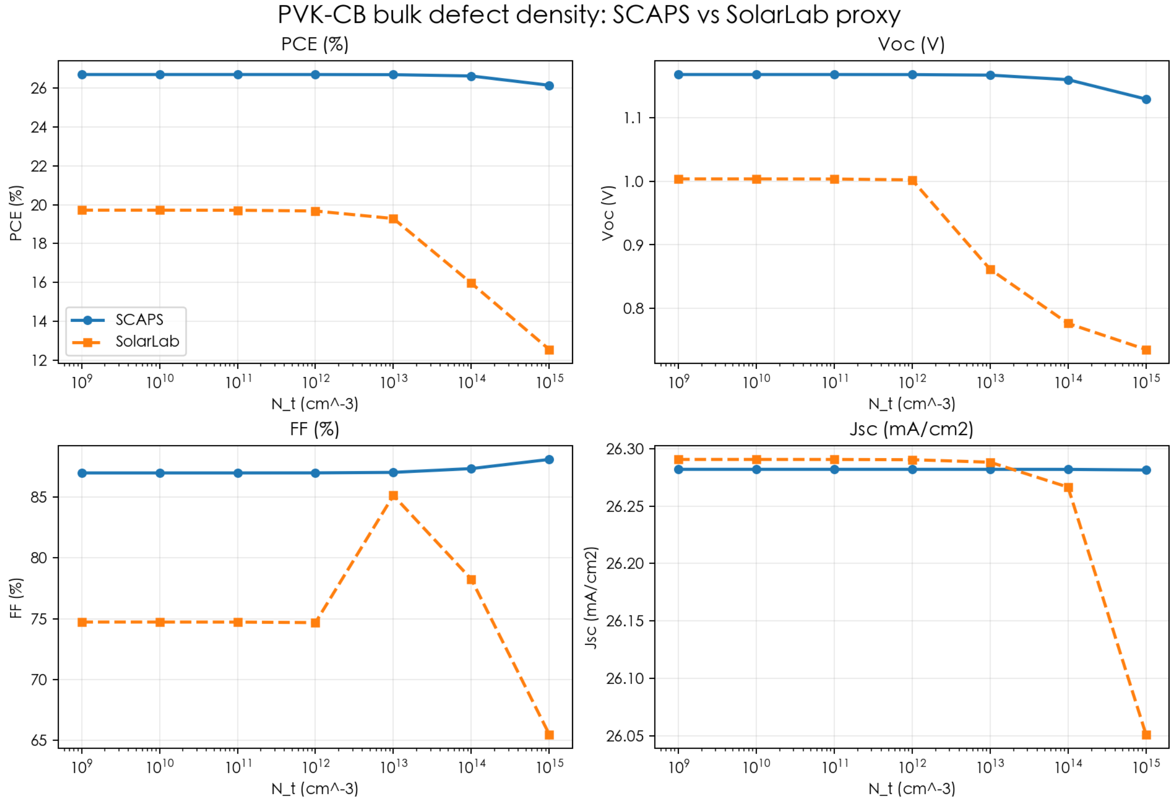
• SolarLab can test this as interface SRV or edge trap profile, but the current sparse run focused on PVK/ETL because it is the sensitive interface.

• Recommended next step: add explicit target-interface trap mapping so HTL/PVK and PVK/ETL can be separated exactly.

PVK-CB bulk defect density scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK-CB bulk defect density scan	1e12 cm^-3	1e9 to 1e15 cm^-3	Perovskite-CB defect density only	absorber SRH lifetime scaled inversely with N_t	Already ran proxy

Visualization



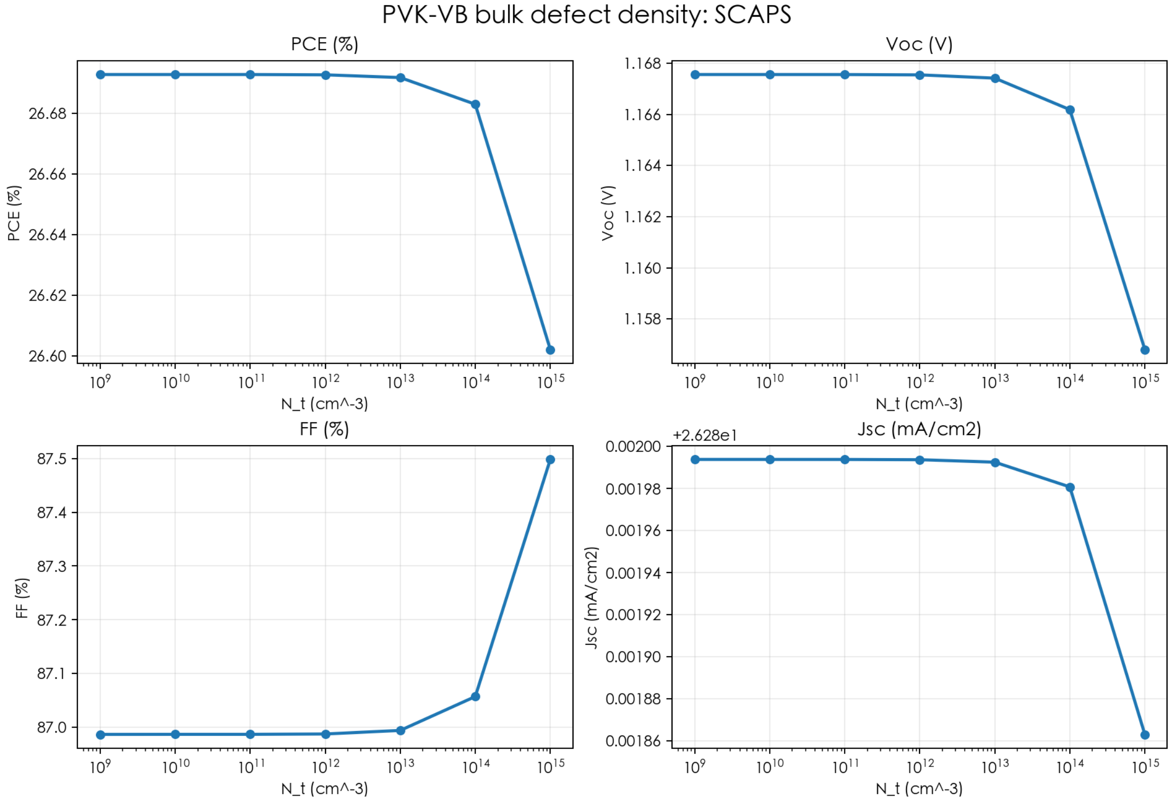
Conclusion

- SCAPS: increasing PVK-CB defect density lowers Voc/PCE moderately.
- SolarLab proxy: increasing bulk N_t lowers Voc/PCE strongly at high density.
- Trend is physically consistent, but exact values depend on capture cross sections and lifetime calibration.

PVK-VB bulk defect density scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK-VB bulk defect density scan	1e12 cm^-3	1e9 to 1e15 cm^-3	Perovskite-VB defect density only	same SRH lifetime proxy is possible; not separately calibrated	Can run; not separately calibrated

Visualization



Conclusion

- SCAPS shows PVK-VB density is less sensitive than PVK-CB.
- SolarLab can represent a VB-like trap by changing n1/p1 reference and lifetime, but the current run did not separately calibrate density response for VB defects.
- This is lower priority than PVK/ETL and CBO because SCAPS sensitivity is weak.

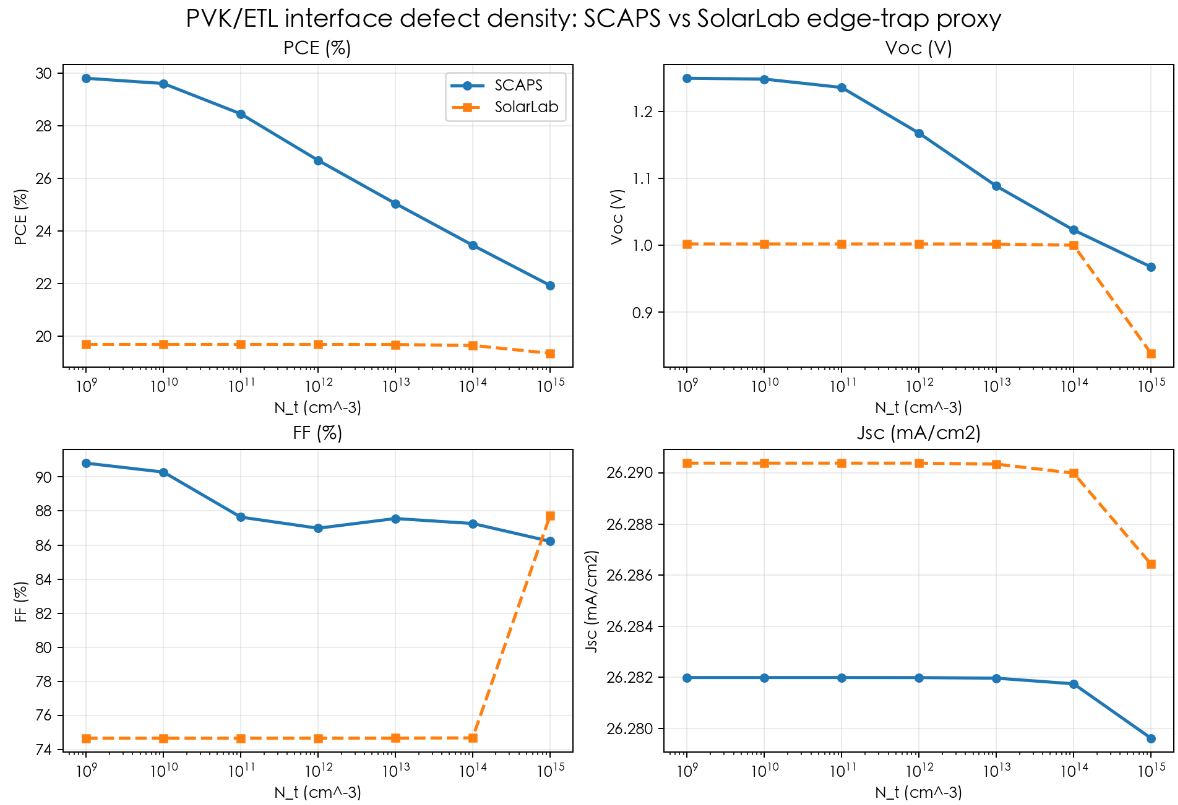
PVK/ETL interface defect density scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK/ETL interface defect density scan	1e12 cm^-3	1e9 to 1e15 cm^-3	Perovskite/ETL defect density only; Gaussian peak scaled with total density	absorber edge trap profile near interfaces; can also use interface SRV	Already ran proxy

Quantitative trend summary

Metric	n	corr	SCAPS range	SolarLab range
PCE (%)	7	0.7	21.93-29.82	19.34-19.67
Voc (V)	7	0.666	0.9678-1.249	0.8388-1.002
FF (%)	7	-0.48	86.22-90.8	74.68-87.72
Jsc (mA/cm2)	7	1	26.28-26.28	26.29-26.29

Visualization



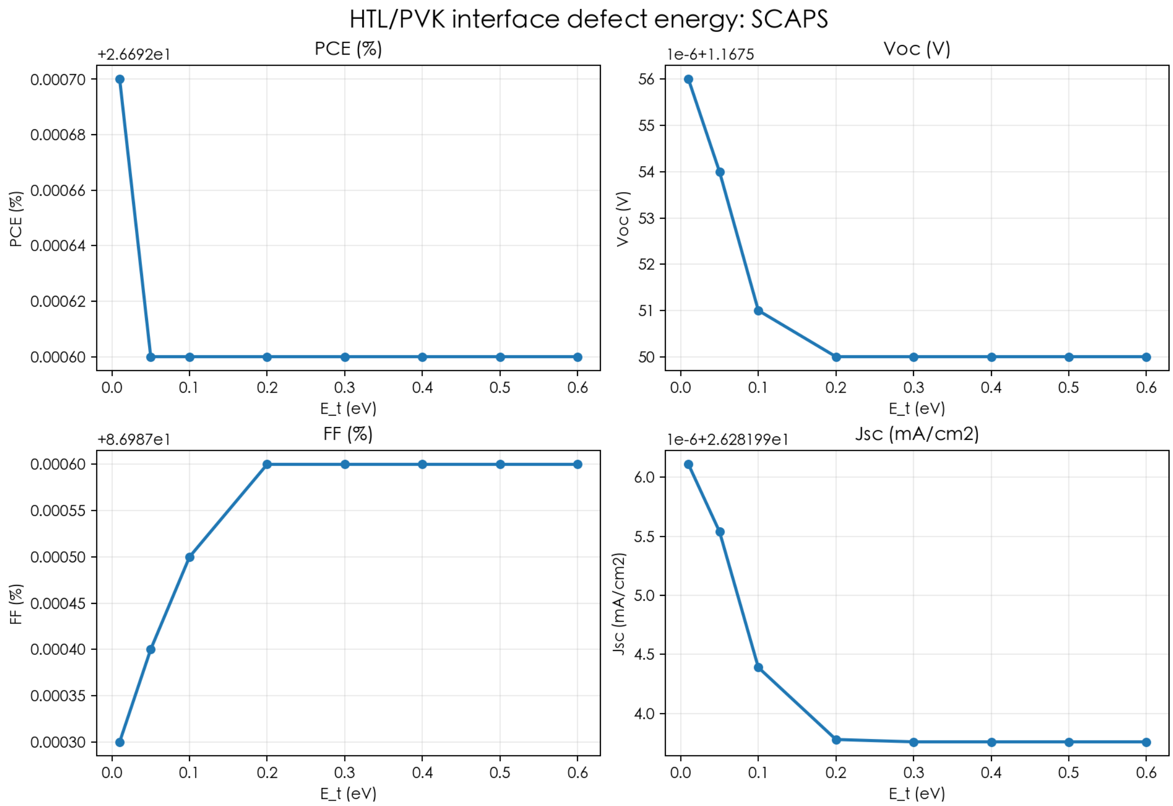
Conclusion

- SCAPS: PVK/ETL is the most sensitive defect-density location.
- SolarLab edge-trap proxy captures high-density Voc/PCE degradation, but the curve is flatter until very high N_t.
- For strict comparison, SolarLab should add per-interface trap density and trap energy rather than symmetric absorber edge profile.

HTL/PVK interface defect energy scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
HTL/PVK interface defect energy scan	site-dependent	0.01 to 0.6 eV	fixed N_t=1e12; change HTL/PVK Et	can map through interface SRV n1/p1, not yet target-interface exact	Can run; not yet strict

Visualization



Conclusion

- SCAPS shows almost no sensitivity for HTL/PVK Et.
- SolarLab can run an equivalent proxy, but it is not the limiting validation case.
- This can be included in a full matrix after target-interface defect mapping exists.

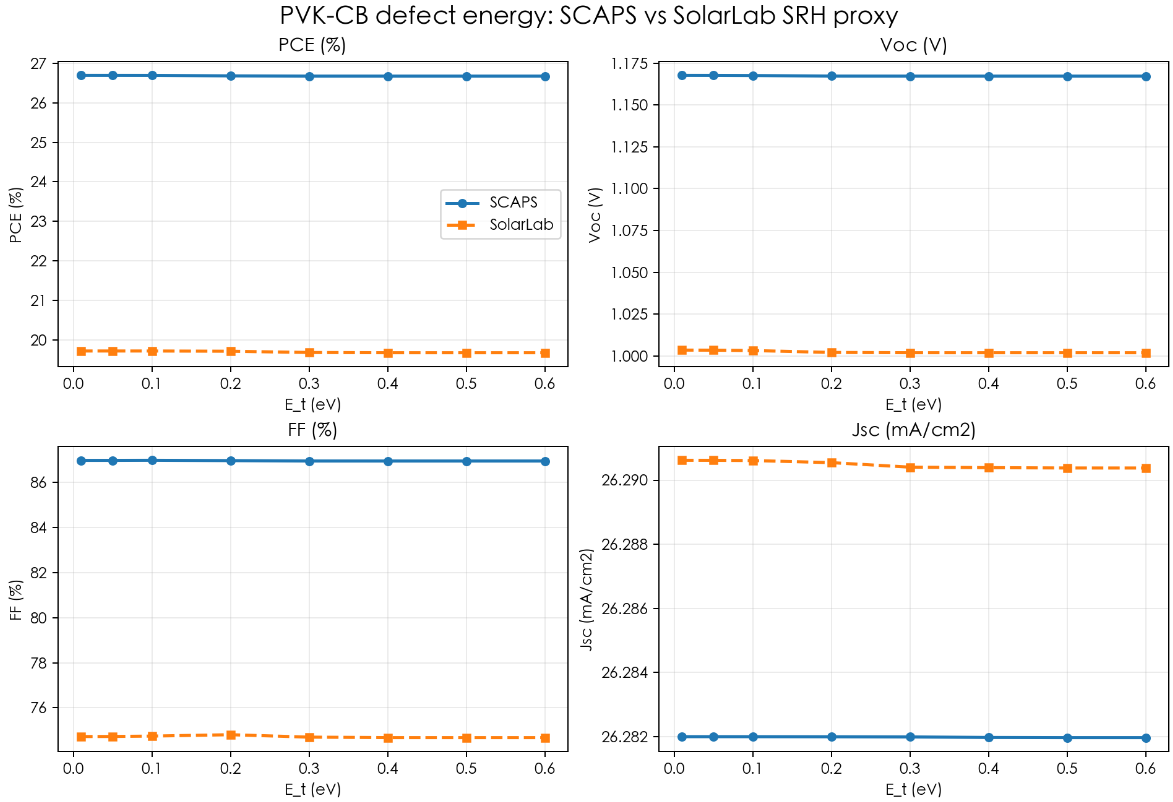
PVK-CB bulk defect energy scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK-CB bulk defect energy scan	site-dependent	0.01 to 0.6 eV	fixed N_t=1e12; change Perovskite-CB Et	absorber SRH n1/p1 from trap depth below CB	Already ran proxy

Quantitative trend summary

Metric	n	corr	SCAPS range	SolarLab range
PCE (%)	8	0.74	26.69-27.73	19.67-19.72
Voc (V)	8	0.898	1.167-1.202	1.002-1.004
FF (%)	8	0.273	86.98-87.8	74.68-74.81
Jsc (mA/cm2)	8	0.911	26.28-26.28	26.29-26.29

Visualization



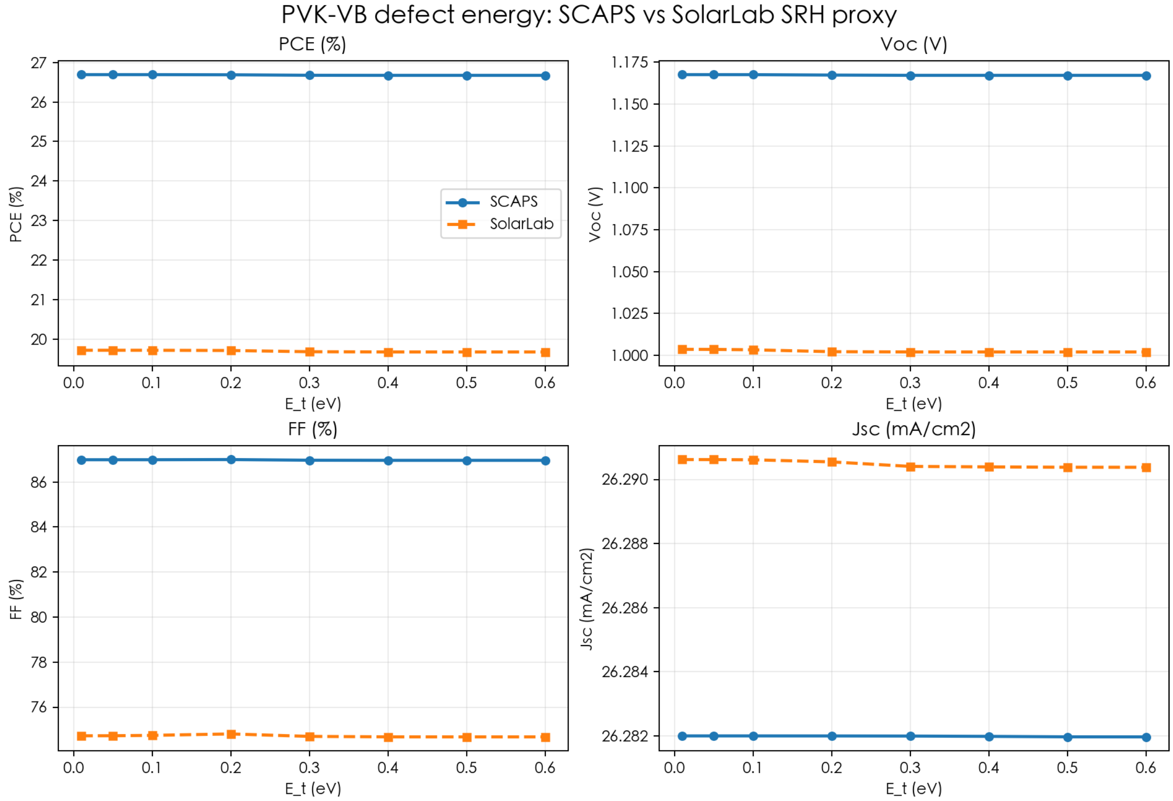
Conclusion

- SCAPS shows shallow-to-deeper CB defect energy causes small PCE/Voc reduction then saturates.
- SolarLab SRH proxy gives the same weak/saturating behavior, although absolute values differ.
- This is a reasonable qualitative match.

PVK-VB bulk defect energy scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK-VB bulk defect energy scan	site-dependent	0.01 to 0.6 eV	fixed N_t=1e12; change Perovskite-VB Et	absorber SRH n1/p1 from trap depth above VB	Already ran proxy

Visualization



Conclusion

• SCAPS sensitivity is weak and saturating.

• SolarLab proxy is also weak and saturating.

• The CB/VB proxy curves are similar because the current base stack is not separately calibrated for asymmetric capture cross sections.

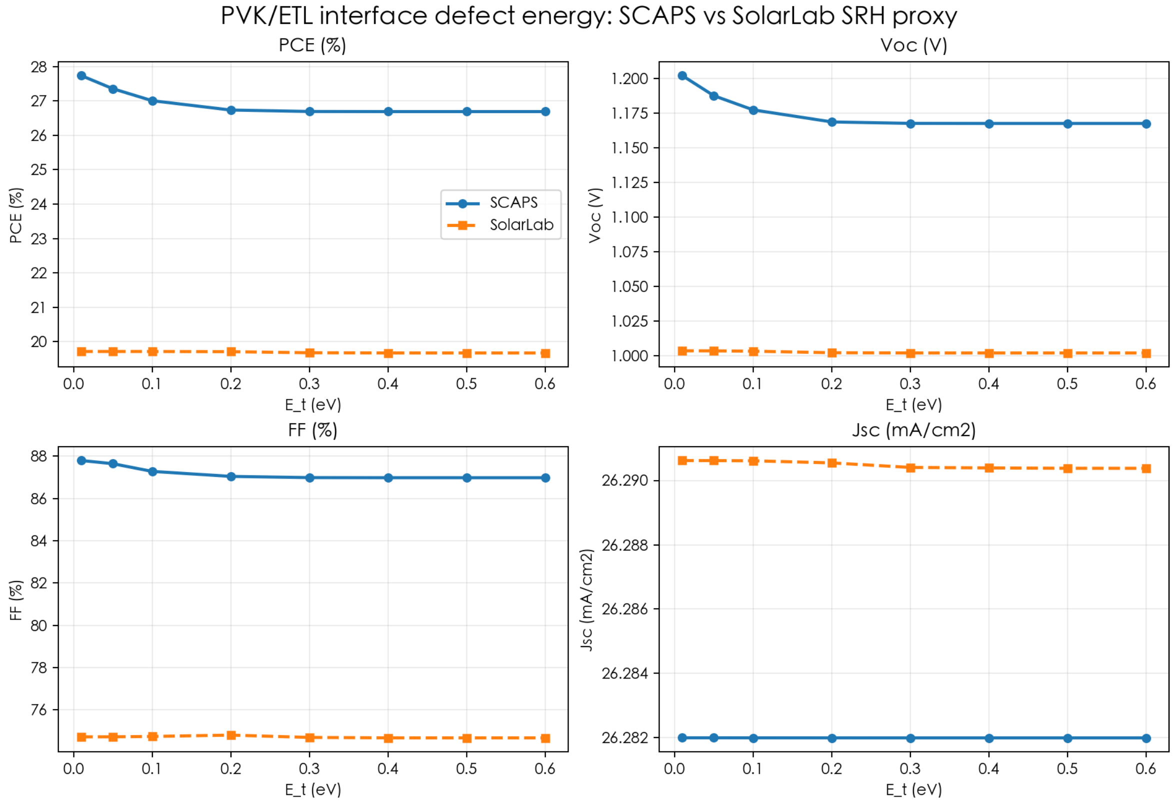
PVK/ETL interface defect energy scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK/ETL interface defect energy scan	site-dependent	0.01 to 0.6 eV	fixed N_t=1e12; change PVK/ETL Et	currently approximated with absorber SRH n1/p1; exact interface Et not implemented	Already ran proxy, exact pending

Quantitative trend summary

Metric	n	corr	SCAPS range	SolarLab range
PCE (%)	8	0.74	26.69-27.73	19.67-19.72
Voc (V)	8	0.898	1.167-1.202	1.002-1.004
FF (%)	8	0.273	86.98-87.8	74.68-74.81
Jsc (mA/cm2)	8	0.911	26.28-26.28	26.29-26.29

Visualization



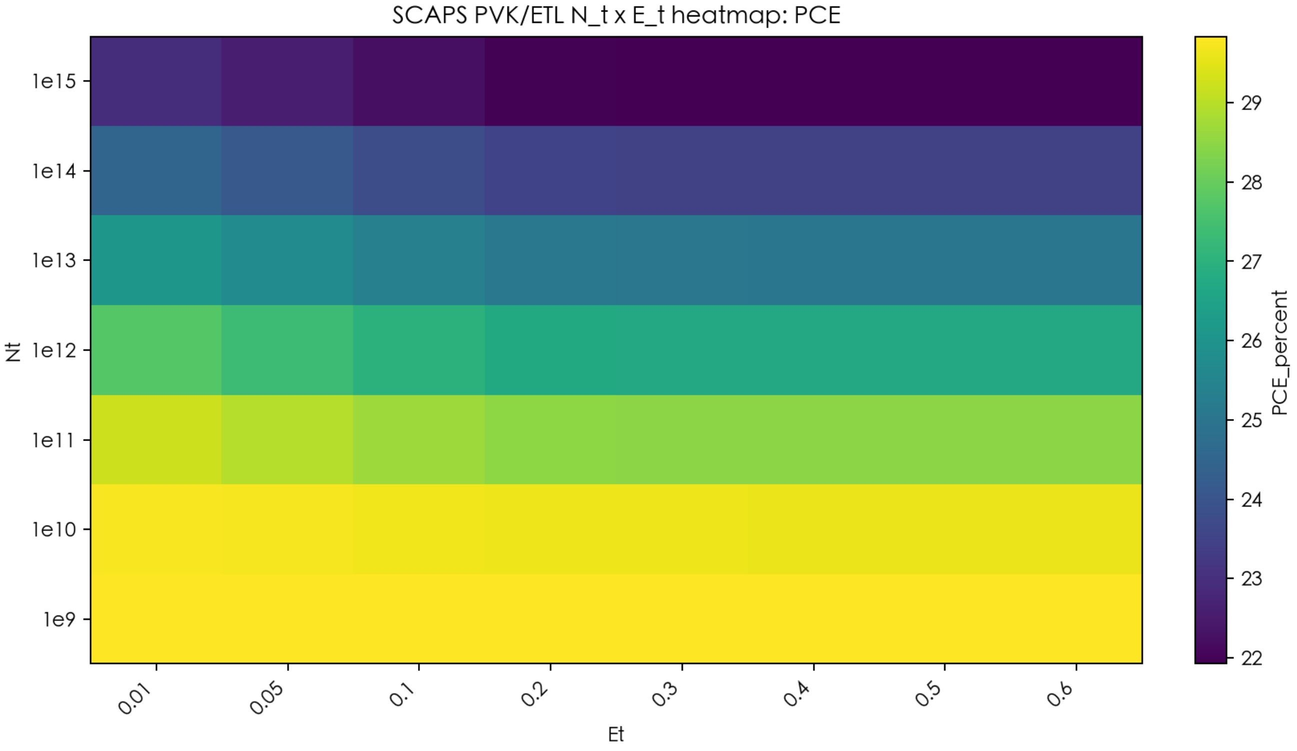
Conclusion

- SCAPS shows shallow PVK/ETL defects produce higher performance; after about 0.3 eV response saturates near base.
- SolarLab proxy captures weak saturation but not the absolute boost.
- Strict interface-energy parity requires interface-specific n1/p1 or explicit SCAPS-style defect object.

PVK/ETL N_t x E_t two-dimensional scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK/ETL N_t x E_t two-dimensional scan	N_t=1e12, Et=0.3 eV	N_t 1e9-1e15; Et 0.01-0.6	SCAPS coupled matrix	SolarLab runner can create the matrix; exact interface-defect mapping pending	SCAPS done; SolarLab matrix not yet full

Visualization



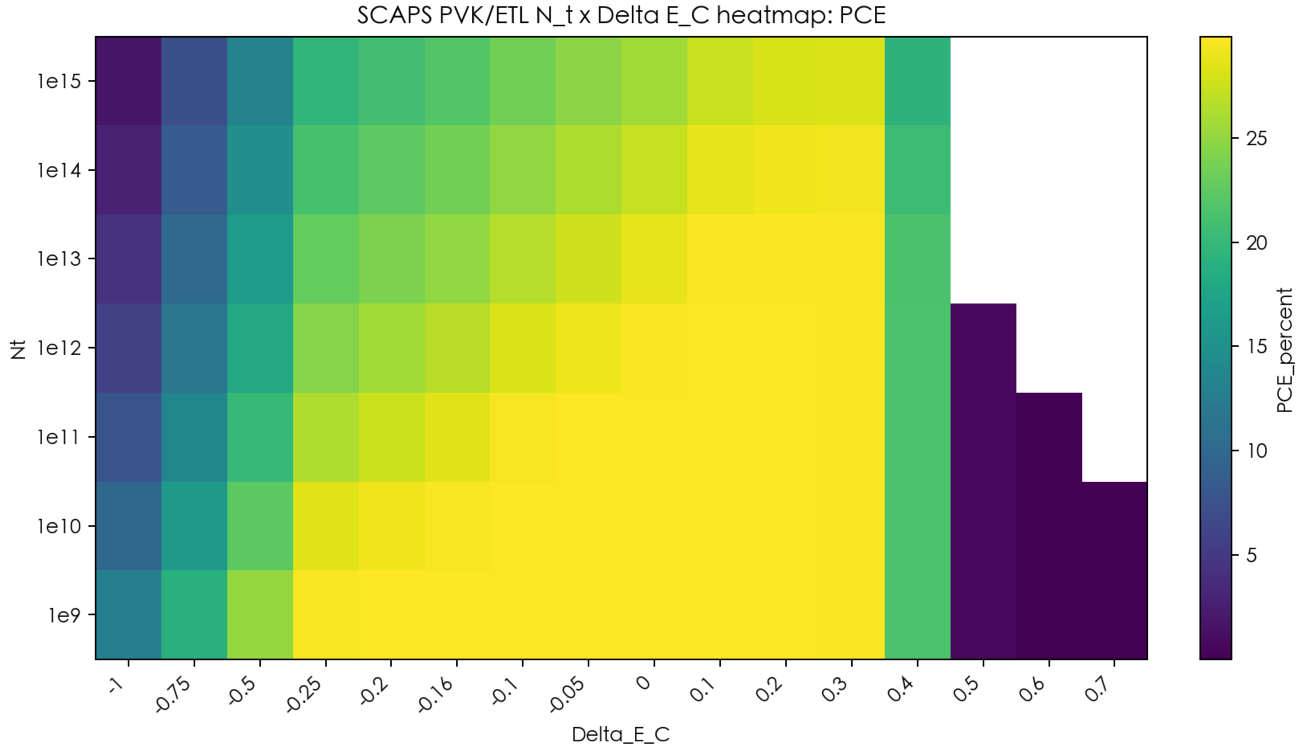
Conclusion

- SCAPS best point: N_t=1e9 cm^-3, Et=0.01 eV, PCE=29.8336%.
- The heatmap is dominated by N_t; E_t matters mainly at high defect density.
- SolarLab can run this as a proxy matrix now, but exact comparison should wait for interface-specific defect mapping.

PVK/ETL N_t x DeltaEc two-dimensional scan

Parameter	Base value	Scan range / levels	SCAPS setting	SolarLab mapping	Status
PVK/ETL N_t x DeltaEc two-dimensional scan	N_t=1e12, DeltaEc=-0.16 eV	N_t 1e9-1e15; DeltaEc -1.0-0.70	SCAPS coupled matrix	SolarLab runner can combine DeltaEc and defect/SRV; CBO response needs calibration	SCAPS done; SolarLab matrix not yet full

Visualization



Conclusion

- SCAPS best point: N_t=1e9 cm^-3, DeltaEc=-0.05 eV, PCE=29.8610%.
- The heatmap shows cliff losses at negative DeltaEc and spike/collection collapse at high positive DeltaEc.
- Because SolarLab sparse CBO response is too flat, this full 2D parity should be treated as pending model calibration.

Overall Conclusion

SolarLab can already execute the parameter families through its API/sweep path without changing solver, boundary conditions, or initial-condition algorithms. In this report, 47 sparse SolarLab points plus 8 extra VB trap-energy proxy points ran successfully.

Best current matches are qualitative trend checks for ETL doping, absorber high-doping degradation, bulk/interface defect density, and weak/saturating defect-energy response. The largest gap is the SCAPS conduction-band cliff/spike behavior, which is much stronger than the current SolarLab chi-only mapping.

Recommended development for strict SCAPS parity: add a SCAPS-parameter mapping module with explicit per-interface defect density, trap energy, capture cross sections, Gaussian peak/total-density conversion, and calibrated heterointerface barrier/SRV controls. Then run full one-factor and 2D matrices with the existing sweep runner.